31/10/25 # Convergence of RVs

→ Sequence of RVs converging to some constant

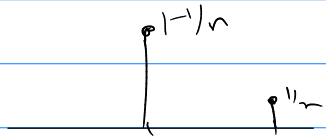
$X_1, X_2, \ldots, X_n, \ldots$

$\lim_{n\to\infty} Pr(|X_n - b| > \varepsilon) = 0$

Sure $\varepsilon > 0$ if $Pr(|X_n - b| > \varepsilon) \to 0$,

then X_n converges to b "in probability"

Ex: (1) $X_n \sim \{0, 1\}$ with $1-1/n$, $1/n$

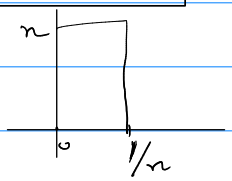


$\varepsilon > 0$

$Pr(|X_n - 0| > \varepsilon) = 1/n \to 0$

$X_n \xrightarrow{p} 0$

(2) $X_n \sim \text{Uniform}[0, 1/n]$



$\varepsilon > 0$

$$Pr(|X_n - 0| > \varepsilon) = \begin{cases} 0 & \varepsilon > 1/n \\ 1 - n\varepsilon & \varepsilon < 1/n \end{cases}$$

$\lim_{n\to\infty} Pr(|X_n| > \varepsilon) = 0$

$X_n \xrightarrow{p} 0$

(3) $X_n \sim Exp(n)$ PDF ne^{-nx}

$$Pr(X_n > \varepsilon) = 1 - \int_0^{\varepsilon} ne^{-nx}\,dx = 1 - (1 - e^{-n\varepsilon}) = e^{-n\varepsilon} \to 0$$

$X_n \xrightarrow{p} 0$

Weak Law of Large Numbers (WLLN)

(SLLN)

$X_1, X_2, \ldots, X_n, \ldots \sim$ iid X $\quad \mu_X < \infty$, $\sigma_X^2 < \infty$

$$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n} \xrightarrow{p} \mu_X$$

Pf:

$$\Pr\left(|\bar{X}_n - \mu_X| > \varepsilon\right) \leq \frac{\sigma_X^2/n}{\varepsilon^2} = \frac{\sigma_X^2}{\varepsilon^2 n} \to 0$$

$E[\bar{X}_n] = \mu_X$

$\mathrm{Var}(\bar{X}_n) = \frac{\sigma_X^2}{n}$

∎

Extension

$X_1, X_2, \ldots, X_n, \ldots$

Y : another RV

$\varepsilon > 0$

$$\Pr\left(|X_n - Y| > \varepsilon\right) \to 0$$

Then $X_n \xrightarrow{p} Y$

Ex: ① $X_n = Y\left(1 + \frac{1}{n}\right)$ $\quad Y \sim \text{Bernoulli}(p)$

$$\Pr\left(|X_n - Y| > \varepsilon\right) \to 0$$

$|X_n - Y| \sim \{0, 1/n\}$ with probabilities $1-p$, p

② $X_1, \ldots, X_n, \ldots \sim$ iid Bernoulli$(1/2)$

$Y \sim$ Bernoulli$(1/2)$ indep. of all X_n

Y \ X_n	0	1
0	1/4	1/4
1	1/4	1/4

$|X_n - Y| \sim \{0, 1\}$ with probabilities $1/2$, $1/2$

$\Pr\left(|X_n - Y| > \varepsilon\right) = 1/2 \quad 0 < \varepsilon < 1$

Convergence in distribution

$X_1, \dots, X_n, \dots$

X: RV

If $F_{X_n}(x) \to F_X(x)$ for every x (at which F_X is continuous)

(CDF) "pointwise convergence"

then $X_n \xrightarrow{d} X$ "in distribution".

Ex: $X_n \sim \text{Bernoulli}(1/2)$ i.i.d

Y: Bernoulli$(1/2)$

$\boxed{X_n \xrightarrow{d} Y}$

→ If $X_n \xrightarrow{p} X$, then $X_n \xrightarrow{d} X$.

→ X_n: discrete $p_{X_n}(k) \to p_X(k) \Rightarrow X_n \xrightarrow{d} X$

→ X_n: mGF $M_{X_n}(t)$ $M_{X_n}(t) \to M_X(t) \Rightarrow X_n \xrightarrow{d} X$

pointwise

Ex: $X_n \sim \text{Binomial}\left(n, \frac{\lambda}{n}\right)$ $X \sim \text{Poisson}(\lambda)$

$\boxed{X_n \xrightarrow{d} X}$

$$p_{X_n}(k) = \binom{n}{k}\left(\frac{\lambda}{n}\right)^k\left(1-\frac{\lambda}{n}\right)^{n-k}$$

$$= \frac{n(n-1)\cdots(n-k+1)}{n^k\, k!}\cdot\lambda^k\,\frac{\left(1-\frac{\lambda}{n}\right)^n}{\left(1-\frac{\lambda}{n}\right)^k}$$

$$= \frac{\lambda^k}{k!}\cdot\frac{\left(1-\frac{1}{n}\right)\cdots\left(1-\frac{(k-1)}{n}\right)}{\left(1-\frac{\lambda}{n}\right)^k}\left(1-\frac{\lambda}{n}\right)^n \to \frac{e^{-\lambda}\lambda^k}{k!}$$

$= p_X(k)$

Central Limit Theorem

Ex: $X_1, \dots, X_n, \dots \sim iid \{-1, 1\}$ with $\frac{1}{2}, \frac{1}{2}$

$Y_n = X_1 + X_2 + \cdots + X_n$

MGF: $\frac{e^{-t} + e^{t}}{2} = 1 + \frac{t^2}{2} + \frac{t^4}{4!} + \cdots$

$M_{Y_n}(t) = \left(\frac{e^{-t} + e^{t}}{2}\right)^n$ no convergence

$\bar{X}_n = \frac{X_1 + \cdots + X_n}{n}$ $\quad M_{\bar{X}_n}(t) = \left(\frac{e^{-t/n} + e^{t/n}}{2}\right)^n = \left(1 + \frac{t^2}{2n^2} + \frac{t^4}{4! n^4} + \cdots\right)^n$

$\bar{X}_n \xrightarrow{P} 0 \Rightarrow \bar{X}_n \xrightarrow{d} 0 \Rightarrow M_{\bar{X}_n}(t) \longrightarrow 1$

$W_n = \frac{X_1 + \cdots + X_n}{\sqrt{n}}$ $\quad M_{W_n}(t) = \left(\frac{e^{-t/\sqrt{n}} + e^{t/\sqrt{n}}}{2}\right)^n = \left(1 + \frac{t^2}{2n} + \cdots\right)^n$

$W_n \xrightarrow{d} N(0,1)$ $\quad \longrightarrow e^{t^2/2}$